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EDUCATION

University of Southern California: Viterbi School of Engineering & Marshall School of Business

Expected May 2027

B.S. Computer Science/Business Administration; Minor in Artificial Intelligence Applications

- GPA: 3.9 • Presidential Scholar • Viterbi Scholar • Dean's List • W.V.T. Rusch Honors • Thematic Option Honors
- Relevant Coursework: Data Structures, Algorithms, Software Engineering, Discrete Methods; Embedded Systems and AI/ML at Bristol, UK

EXPERIENCE

Software Engineer Intern, Matha Labs

Jun 2026 – Present

- Replaced AWS Cognito's hosted UI on humanhud.ai (Next.js/TypeScript, static export on S3 and CloudFront) with a branded in-site sign-in: choice-based USER_AUTH covering password and email one-time-code factors, account confirmation and reset, plus Google SSO over browser PKCE (S256) with callback-state validation and same-site redirect sanitization.
- Shipped the signed-in dashboard and its five action commands (add or scan a receipt, add or scan a nutrition label, log weight) writing to the same stores the live HUD reads, with opt-in on-device OCR and a confirm step before logging.
- Contributed to the SwiftUI iOS app for Meta Ray-Ban Display glasses, which renders a two-slot health and money HUD over Meta's Wearables Device Access Toolkit and feeds it HealthKit vitals, Vision OCR scans, and Plaid balances via a Lambda/DynamoDB backend.
- Wired Stripe payments, AWS Cognito accounts, and a correctness pass over the core trading logic for Prosperity (joinprosperity.com), a copy-trading platform, and authored its complete Alpaca brokerage integration documentation for certification review.

AI Engineer, USC Industry Collaboration Program

Sep 2026 – Present

- Only AI engineer on a cross-functional student team building internal software for a Canadian construction company (details under NDA).
- Own the AI engineering for automating a core internal workflow, from requirements to delivery.

Software Lead, USC RoboSub (Barracuda AUV)

Jan 2025 – Present

- Promoted three times in 18 months (Software Engineer → Perception Lead → Perception & Autonomy Lead → Software Lead); now direct 5–10 engineers across the AUV's vision, SLAM, planning, and mission-control stack (ROS 2, NVIDIA Jetson).
- Migrated the team's inherited 10,000+ line ROS 1 codebase to ROS 2, auditing undocumented legacy nodes and cutting dead code in an architecture redesign; the AUV now runs entirely on the new stack.
- More than doubled underwater detection accuracy (0.40 → 0.90 mAP) by building the dataset and training YOLO and RT-DETR, then compiled to ONNX → TensorRT (FP16/INT8) to hit real-time inference on-vehicle.
- Delivered the vehicle's SLAM capability, fusing stereo (ZED) landmarks, IMU, DVL, and depth into a GTSAM factor graph to drive costmap navigation; validated on rosbag replay against surveyed prop placement, holding every landmark to under 0.3 m absolute error.
- Root-caused the AUV's sluggish surge response from 9 wet-test rosbags (75 GB): a thrust clamp misconfigured at 12 N instead of 30 N; drove the fix into the team's control config and retuned the GPU-parallel MPPI controller (CUDA/C++).

Director of Digital Strategy, USC Undergraduate Student Government

May 2025 – Present

- Own USG's digital infrastructure serving 20,000+ students, shipping front-end features on the official WordPress site (dynamic announcements, interactive election interfaces); invited back for a second year.
- Made \$352K in student-government spending publicly searchable across 40+ organizations, assemblies, and committees, cutting manual reporting 95% with automated Senate Bill, Legislative Project, and Funding trackers.

TECHNICAL PROJECTS

Ask USG: RAG Chatbot for USC Student Government

Node/Express • Python • OpenAI • pgvector • LangGraph • Docker

- Built a production RAG chatbot for USC's 20,000+ undergraduates, unifying three source types on a weekly GitHub Actions schedule: 97 auto-discovered site pages, 2 multi-tab Google Sheets (1,000+ rows), and a live Google Calendar; campus-wide launch Fall 2026.
- Improved answer quality 83% → 95% by turning 1,100 real beta questions from 80+ USG officers into an automated eval pipeline, then fixing retrieval completeness (sibling-chunk expansion), hallucination guards, and live-calendar grounding.
- Replaced pure vector search with hybrid retrieval (BM25 fused with pgvector similarity) and cut time-to-first-token 92% (6.2 s → 0.5 s) via SSE streaming and eliminating hosting cold starts.
- Extended it into a LangGraph agent routing between retrieval and live data tools, with guardrails that hand crisis and academic-integrity requests to human support, and hardened it against prompt injection (zero hijacks across an 11-probe adversarial battery).

PTDrone: Drone Ground Control & Manufacturing Cell (Capstone)

Next.js • React • Render

- Lead a five-person CSCI 401 team building, for an outside stakeholder, a drone ground control station, an AI chatbot that answers parts questions and triggers on-demand manufacturing, and a simulated manufacturing cell.
- Set up the production web pipeline: Next.js static export on Render, with preview-branch review gating deploys to ptdrone.tech.

SKILLS

Languages: Python, C++, CUDA, Java, JavaScript, TypeScript, Swift, C, MATLAB, HTML/CSS

AI, Robotics & ML: OpenAI API, LangGraph, Langfuse, RAG, ROS 2, PyTorch (YOLO, RT-DETR), TensorRT, ONNX, GTSAM, NVIDIA Isaac Sim

Web & Infrastructure: React, Next.js, Node/Express, SwiftUI, PostgreSQL (pgvector, full-text search), AWS (Cognito, Lambda, S3/CloudFront, DynamoDB), Stripe, WordPress, Render, Docker, K3s (Kubernetes), unit & integration testing, GitHub Actions (CI/CD), Git, Linux/Bash